

# *B.N.M. Institute of Technology*

**An Autonomous Institution under VTU**

Approved by AICTE, Accredited as Grade A Institution by NAAC till 31.12.2026.

All Eligible UG branches – CSE, ECE, EEE, ISE & Mech. Engg. Accredited by NBA till 30.06.2025

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Vidyayāmruthamashnuthe

## **INNOVATIVE PROJECT LAB 1 (21ECE139)**

**(Innovative Project Lab Course)**

**PROJECT REPORT**

**ON**

**“LPG ASSISTANT”**

**by**

**Sanjana HS**

**1BG21EC080**

**Thanishka K**

**1BG21EC098**

**Ujwal BS**

**1BG21EC101**

Under the guidance of

**Sri.Vyasraj T**

Assistant Professor

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## **Department of Electronics and Communication Engineering**



Vidyayāmṛthamashnute

## **CERTIFICATE**

Certified that the Innovative Project titled “LPG ASSISTANT” carried out by **Sanjana H.S(1BG21EC080)**, **Thanishka K(1BG21EC098)**, and **Ujwal B.S(1BG21EC101)** , bona fide students of **III semester**, during the year **2022-2023**, for the fulfillment of the academic requirements for the IPL Course **Innovative Project Lab -1 (21ECE139)**.

The Project report has been approved as it satisfies the academic requirements in respect to the course.

**Dr. Rekha P**

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BNMIT, Bangalore

**Dr. P. A. Vijaya**

Professor and Head

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# **CHAPTER 1**

## **INTRODUCTION**

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# CHAPTER 1

## INTRODUCTION

### 1.1 Introduction

Liquefied petroleum gas (LPG) is a by-product of natural gas extraction and crude oil refining. It is a mixture of hydrocarbon gases, the most common being butane and propane. At room temperature, LPG is a colourless and odourless non-toxic flammable gas commonly used in residential heating, cooking and hot water systems. It is an energy-rich fuel, with greater caloric value (heating value) compared to natural gas, coal, or other common fuels, is a highly convenient fuel source and is also easy to store. It is cleaner than most other fossil fuels, with low sulphur content, low emissions of black carbon, and no spills. Its use helps keep our air clean and emissions of greenhouse gases low compared to use of other fossil fuels. Gas for cooking is one of the most popular uses for LPG in homes. Gas cooktops and ovens are the most common uses, but outdoor gas BBQs are also seen in most places. LPG is the preferred fuel for home use for its Efficiency, Portability and Cleaner Power. Though it is the most basic necessity we tend to forget to refill or replace it and it becomes very problematic when it empties in the middle of use.

Gas leakage is a critical safety concern in households and industries that deal with combustible gases like LPG. The leakage of gas can lead to fire, explosions, and even death. To prevent such accidents, a gas leakage detection and gas quantity monitoring system can be developed using NodeMCU ESP32, integrating a gas sensor and load cell.

This system can detect gas cylinder leakages and inform the user with real-time gas level notifications. This system can also monitor the gas quantity and provide the user with information about the remaining gas level in the cylinder.

### 1.2 Motivation

Gas leakage is a serious safety concern that can lead to fire, explosions, and even death. A gas leakage detection and gas quantity monitoring system can help prevent such accidents by alerting the user about the gas leakage and monitoring the remaining gas level in the cylinder. This system can provide peace of mind to the users and help prevent accidents.

### 1.3 Problem Statement

The problem statement is to develop a gas leakage detection and gas quantity monitoring system using NodeMCU ESP32, integrating gas sensor and load cell. The system should detect gas cylinder leakages and inform the user with real-time gas level notifications. It should also monitor the gas quantity and provide the user with information about the remaining gas level in the cylinder.

### 1.4 Objective

The aim of this proposed system is to monitor the LPG in household applications ,provide insight on different precautions to be taken and give safety instructions in case of mishaps and leakage detection. The system shows the remaining time of the LPG cylinder can be used in full flame. The gas sensor is used to detect the amount of LPG in the atmosphere and alerts the consumer about the gas leakage if any by sending SMS to a particular mobile number. Nowadays the booking of the LPG cylinders is by sending SMS to the booking agent. In this system the Mcu module automatically books the cylinder by SMS to the agent booking number. This will help the owner to get the new cylinders in time without any fail.

The objective of this project is to develop a gas leakage detection and gas quantity monitoring system using NodeMCU ESP32, integrating gas sensor and load cell. This system aims to detect gas cylinder leakages and inform the user with real-time gas level notifications. It also aims to monitor the gas quantity and provide the user with information about the remaining gas level in the cylinder.

### 1.5 INNOVATION

- We plan to create a website that provides insight on different precautions to be taken and give safety instructions in case of mishaps and leakage detection.
- Add a sensor that detects leakage and alerts the user about it.
- Instructions to reach safety in case of a leak.
- Alert the nearest fire station.

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# **CHAPTER 2**

## **LITERATURE SURVEY**

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## CHAPTER 2

### LITERATURE SURVEY

#### LITERATURE SURVEY

Leakage of gas is a major problem in industries and homes. One of the preventive methods to stop accident associated with the gas leakage is to install a gas leakage detection kit at vulnerable places and to automatically detect and alert the people about the leaking gas. The gas may leak from various places like the top of the cylinder where the pipe carrying the gas to the application is present, or the pipe may get cracked leading to leakages. Hence accidents occur when a spark occurs in a place where this leaked gas is present. Same scenario can be seen in household activities. The working of any advanced system mainly depends on the microcontroller which controls the entire functioning of the device.[1]

Incorrect use of LPG can result in explosions, fires, loss of property and most importantly, loss of lives. If you hear a hissing sound or the olfactory senses are sensing a foul smell, then it is evident the LPG gas is leaking. If the gas leak has already started the instead of panicking here are the measures you can take. Open the door sand windows of your home, so the gas can go out. It will give you ample time to call the professionals from the gas agency. Do not switch on lights or fans. Avoid lighting cigarettes or any appliances near the burner. Do not turn on burners when the gas is leaking. Keep the stove and cylinder off when not in use.[2]

A Safety system for automatic detecting system and gas leakage is proposed. The device is intended for household safety where LPG is used. After giving input, there is no requirement of human intervention and completely working. This work improves the existing safety paradigm used in industry and it may also be applied in homes and workplaces[3].

In this modern society and increasing, Liquid Petroleum Gas (LPG) population usage is more required. The leakage of gas causes a serious hazard in many cases. A Safety system for automatic switching system and gas leakage detection is proposed. The device is intended for household safety where LPG is used. This system contains two modules: the Delay module and the Detection module. The Delay module works based on input and activates the system. The detection module checks change in threshold values of gas (es) level and activate the system. We can add modules like a buzzer for an audiovisual alarm for the system. This system is an advantage compared to the existing system as all are secondary safety systems. After giving input, there is no requirement of human intervention and completely working with the battery model to avoid circuit complexity. By detection and delay we can prevent damage and it won't effect the environment.[4]

A load cell converts a force such as tension, compression, pressure, or torque into an electrical signal that can be measured and standardized. It is a force transducer. As the force applied to the load cell increases, the electrical signal changes proportionally. The most common types of load cell are pneumatic, hydraulic, and strain gauges[5]

Home fires have been taking place frequently, and the threat to human lives and properties is growing. A gas leakage alert system is presented to detect the gas leakage and to alarm the people onboard. Gas leakage leads to various accidents, including explosions, firing, suffocation, and choking. The reason for such outbursts is due to substandard cylinders, old valves, worn-out regulators, and a lack of awareness in handling gas cylinders. A system to detect LPG leaks is needed to avoid fire accidents and provide house safety. Several LPG detection and alert system designs have been proposed in the literature. The LPG leakage detection and alert system presented in this section is simple, yet reliable. It is battery-operated and can also be operated with ac power supply. A gas sensor is used to detect LPG. The sensor can detect the LPG concentration in the range of 200-10000 ppm and the buzzer starts to produce an alarm sound if the gas concentration is greater than the threshold value. Gas leakage leads to severe accidents, resulting in



material losses and human injuries. A simple yet reliable LPG leakage detection and alert system are presented.[6]

A sensor is a technological device that detects/senses a signal, physical condition, and chemical compounds. It is also defined as any device that converts a signal from one form to another. Sensors are mostly electrical or electronic. A gas sensor measures the concentration of gas in its vicinity. It identifies gases by measuring their break down voltages. Each gas has a unique breakdown voltage i.e. the electric field at Which it is ionized. The sensor identifies gases by measuring these voltages. The concentration of the gas can be determined by measuring the current discharge in the device. Gas sensing technologies: Metal Oxide Based Gas Sensors Capacitance Based Gas Sensors Acoustic Wave Based Gas Sensors Calorimetric Gas Sensors Optical gas sensors Electrochemical gas sensors Carbon dioxide (CO<sub>2</sub>) gas sensor users infrared light to measure temperature. The temperature increases due to the infrared light not being absorbed by CO<sub>2</sub>. Hydrogen gas sensor. Mostly palladium is used to detect hydrogen because palladium selectively absorbs hydrogen gas and forms the chemical palladiumhydride.[7]

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# **CHAPTER 3**

## **METHODOLOGY**

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## CHAPTER 3

### METHODOLOGY

#### 3.1 Constuction

The gas leakage detection and gas quantity monitoring system can be developed using the following steps:

##### 1. Calibration of Load Cell

The load cell needs to be calibrated before measuring the weight of the gas cylinder. To calibrate the load cell, connect it to the HX711 Load Cell Amplifier Module and follow the calibration procedure using the Arduino IDE and HX711 library.[13]

##### 2. Connecting the Hardware Components

Connect the MQ-2 gas sensor, load cell, and 16x2 LCD display to the NodeMCU ESP32 using the breadboard and connecting wires.

##### 3. Developing the Software

Develop the software using Arduino IDE and Adafruit libraries. The software should read the gas sensor data and load cell data and display it on the LCD display. It should also send the data to the cloud using the Adafruit IO and MQTT libraries. The cloud can be used to monitor the gas level and detect gas cylinder leakages.

##### 4. Testing the System

Test the system to ensure that it is functioning properly. Test the gas leakage detection by simulating a gas leakage using a gas lighter near the gas sensor. Test the gas quantity monitoring by measuring the weight of the gas cylinder and comparing it with the gas level displayed on the LCD display.

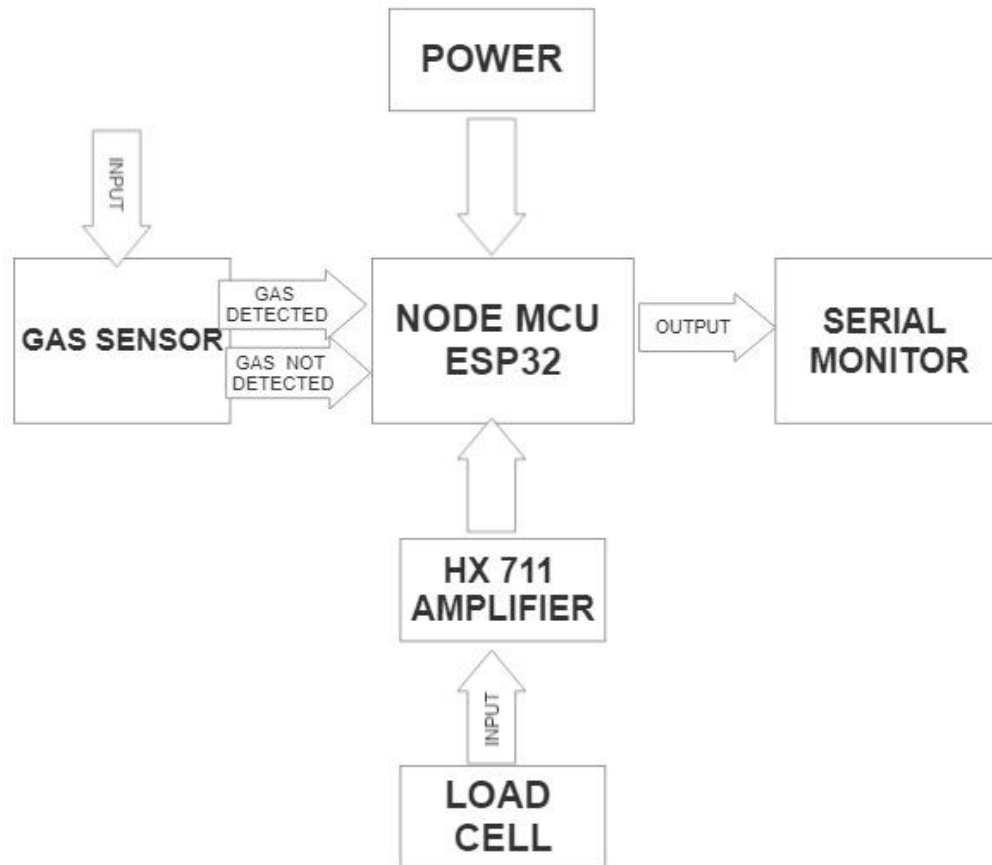


Fig 3.1 Block Diagram

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# **CHAPTER 4**

## **HARDWARE AND SOFTWARE REQUIREMENTS**

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## CHAPTER 4

### HARDWARE AND SOFTWARE

#### 4.1 Hardware Requirements

##### NodeMCU ESP32

NodeMCU ESP32 is a development board based on the ESP32 microcontroller. It is similar to the NodeMCU ESP8266, but with more powerful hardware and additional features. The ESP32 is a low-cost, low-power system-on-chip (SoC) microcontroller with integrated Wi-Fi and Bluetooth capabilities. It is designed for Internet of Things (IoT) applications, including home automation, industrial automation, and wearable devices.

Some of the features of the NodeMCU ESP32 include:

- Dual-core Tensilica LX6 microprocessor
- Integrated Wi-Fi and Bluetooth
- 4 MB flash memory
- 520 KB SRAM
- Multiple digital and analog input/output pins
- Built-in USB-serial converter
- JTAG debugging support
- Support for various communication protocols such as SPI, I2C, UART, and more.

The NodeMCU ESP32 can be programmed using the Arduino IDE or other programming environments, and there are many libraries available for it, making it easy to develop IoT applications. It is also compatible with many sensors and actuators, making it a popular choice for DIY projects and prototyping.[8]

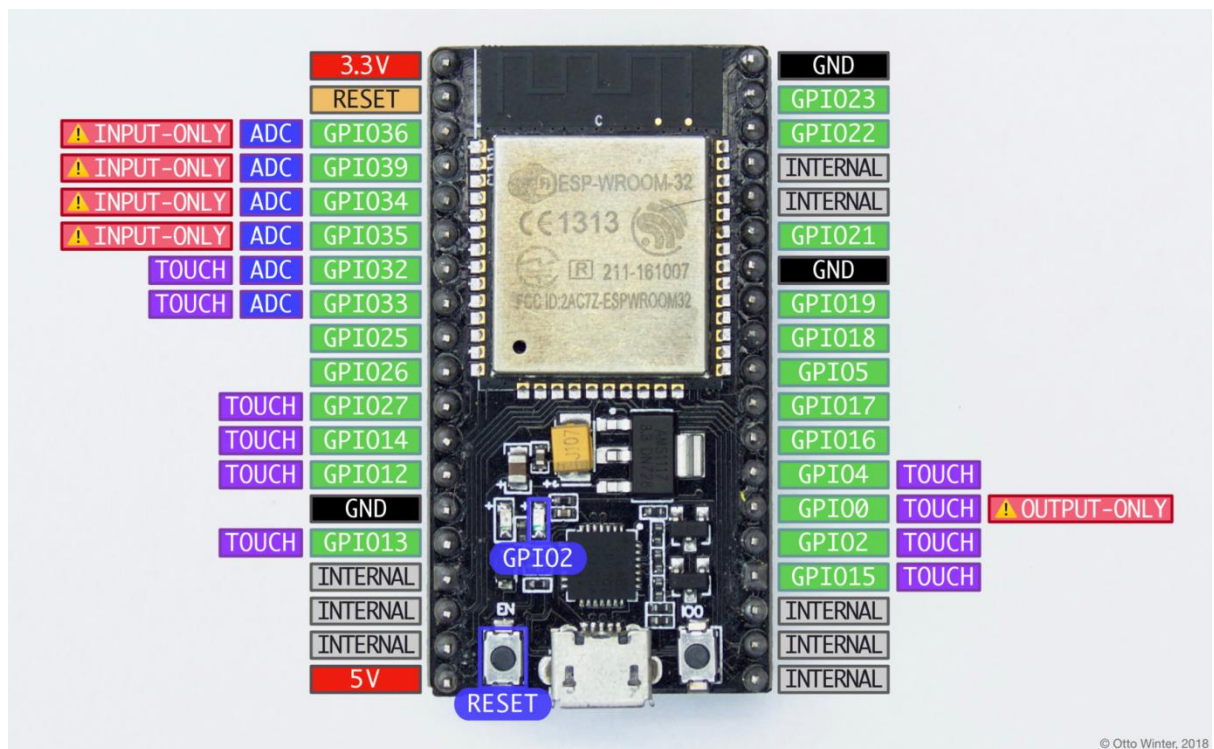


Fig 4.1 Pins on the NodeMCU ESP32 development board.

The ESP32 boards often use the internal GPIO pin numbering on the board, this means that you don't have to worry about other kinds of pin numberings.

Note that in certain conditions you *can* use the pins marked as **INTERNAL** in the above image.

- **GPIO0** is used to determine the boot mode on startup. It should therefore not be pulled LOW on startup to avoid booting into flash mode. You can, however, still use this as an output pin.
- **GPIO34-GPIO39** can not be used as outputs (even though GPIO stands for “general purpose input **output**”...).
- **GPIO32-GPIO39**: These pins can be used with the Analog To Digital Sensor to measure voltages.
- **GPIO2**: This pin is connected to the blue LED on the board as seen in the picture above. It also supports the touch pad binary sensor as do the other pins marked **touch** in the above image.

- 5V is connected to the 5V rail from the USB bus and can be used to power the board. Note that the UART chip is directly connected to this rail and you therefore **cannot** supply other voltages into this pin.

## MQ-2 gas sensor

The MQ2 sensor is one of the most widely used in the MQ sensor series. It is a MOS (Metal Oxide Semiconductor) sensor. Metal oxide sensors are also known as Chemiresistors because sensing is based on the change in resistance of the sensing material when exposed to gasses.

MQ-2 gas sensor is a gas detection sensor module that is commonly used in electronic projects and systems to detect various types of gases, such as smoke, propane, butane, methane, and alcohol. The sensor module is based on the MQ-2 gas sensor chip, which uses a small heater to heat a sensing element, consisting of a tin dioxide ( $\text{SnO}_2$ ) thin film, to detect the target gases.

The MQ-2 gas sensor module can be easily interfaced with microcontrollers and development boards like Arduino, Raspberry Pi, and NodeMCU. It has four pins, namely VCC, GND, Analog Output (AO), and Digital Output (DO). The analog output voltage from the sensor module varies with the concentration of the detected gas, and the digital output pin changes its state when the gas concentration crosses a threshold level, set using a potentiometer on the module.

The MQ-2 gas sensor has a high sensitivity and a fast response time, making it suitable for applications like gas leakage detection, indoor air quality monitoring, and fire detection. However, it is important to note that the sensor may not be accurate in some conditions and may also be affected by environmental factors like temperature and humidity.

The MQ-2 gas sensor module can be calibrated for different gases using a calibration curve, which relates the output voltage of the sensor module to the concentration of the gas. It is also recommended to use the sensor module in a well-ventilated area and to take proper safety precautions when working with potentially hazardous gases.



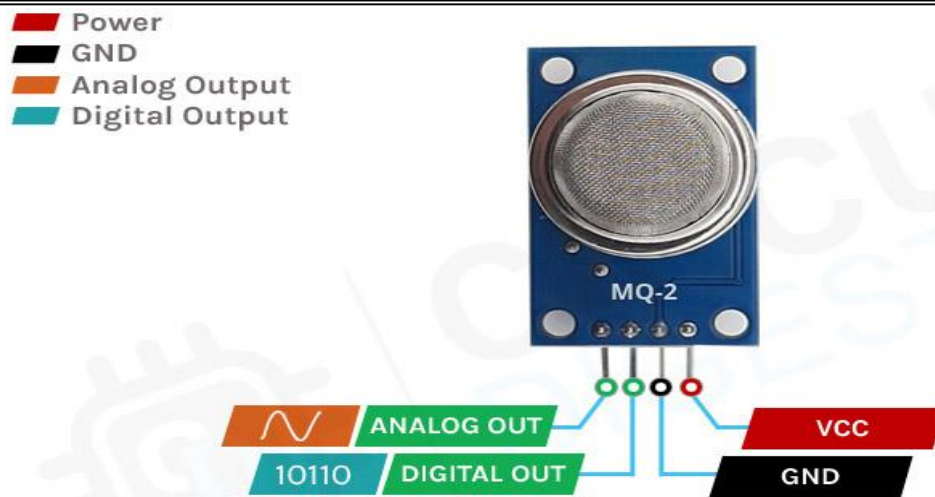


Fig 4.2 MQ-2 Gas sensor pinout

## Internal structure of MQ2 Gas Sensor

The MQ2 is a heater-driven sensor. It is therefore covered with two layers of fine stainless steel mesh known as an “anti-explosion network”. It ensures that the heater element inside the sensor does not cause an explosion because we are sensing flammable gasses.



Fig 4.3 Gas Sensor

It also protects the sensor and filters out suspended particles, allowing only gaseous elements to pass through the chamber. A copper-plated clamping ring secures the mesh to the rest of the body.

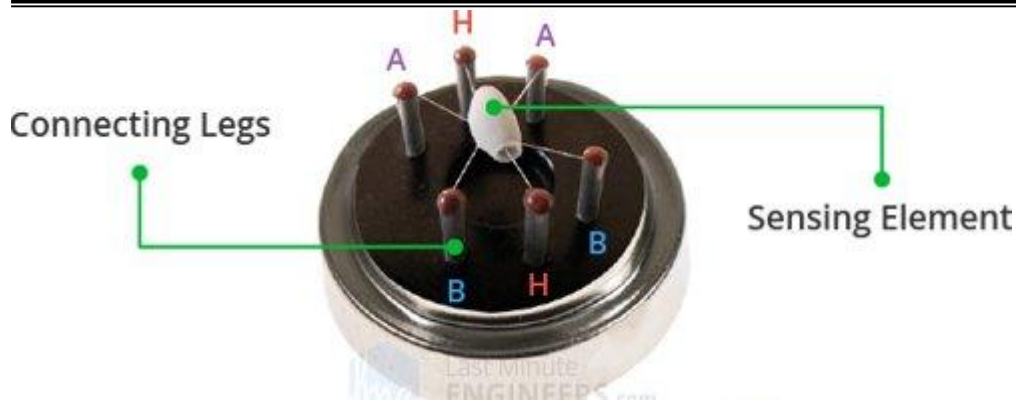


Fig 4.4 Internal structure of MQ2 Gas Sensor

When the outer mesh is removed, the sensor looks like this. The sensing element and six connecting legs that extend beyond the Bakelite base form the star-shaped structure. Two (H) of the six leads are in charge of heating the sensing element and are linked together by a Nickel-Chromium coil (a well-known conductive alloy).

The remaining four signal-carrying leads (A and B) are connected with platinum wires. These wires are connected to the body of the sensing element and convey slight variations in the current flowing through the sensing element.

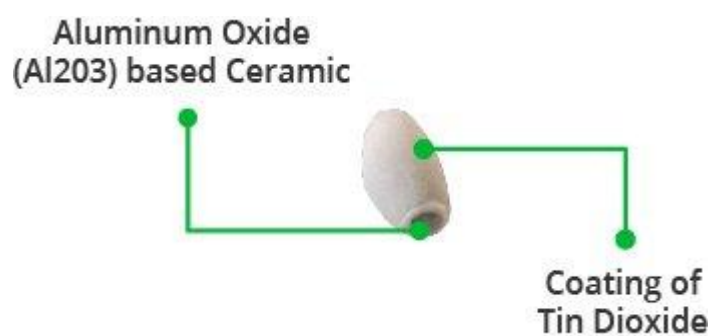


Fig 4.5 Sensing Element

The tubular sensing element is made of Aluminum Oxide ( $\text{Al}_2\text{O}_3$ ) based ceramic with a Tin Dioxide coating ( $\text{SnO}_2$ ). Tin Dioxide is the most important material because it is

sensitive to combustible gasses. The ceramic substrate, on the other hand, improves heating efficiency and ensures that the sensor area is continuously heated to the working temperature.

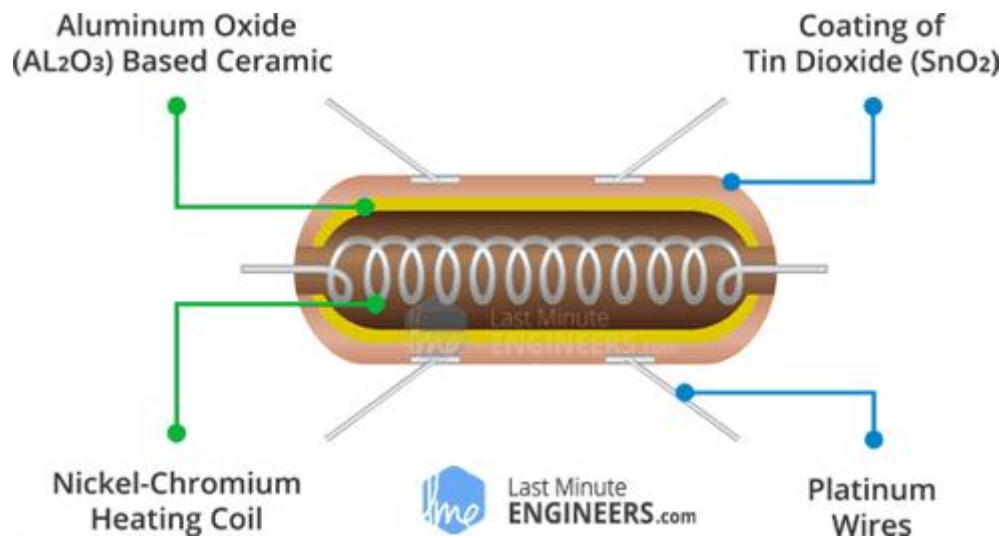


Fig 4.6 Internal structure of sensing element

To summarize, the Heating System is composed of a Nickel-Chromium coil and an Aluminum Oxide-based ceramic, while the Sensing System is composed of Platinum wires and a Tin Dioxide coating.[9]

## Load cell



Fig 4.7 Load cell

A load cell is a type of sensor that is used to measure force or weight. It is commonly used in various applications such as weighing scales, industrial automation, and material

testing. Load cells work on the principle of strain gauge technology, which converts the mechanical force applied to the load cell into an electrical signal.

A load cell consists of a metal structure that is designed to deform when a force is applied to it. The deformation of the metal structure causes a change in the resistance of the strain gauges that are attached to it. The strain gauges are arranged in a Wheatstone bridge configuration and are connected to a signal conditioner circuit that amplifies and processes the output signal.

Load cells are available in different types, such as single point, shear beam, and compression load cells, depending on their application and design. They also come in different weight capacities and accuracy levels.

Load cells are commonly interfaced with microcontrollers or other electronic devices using analog or digital interfaces, such as Analog-to-Digital Converters (ADCs) or serial communication protocols like SPI or I2C. The output signal from the load cell can be used to calculate the weight or force applied to it, and this information can be displayed on an LCD or sent to a computer or other electronic devices for further processing.

Load cells require proper calibration and maintenance to ensure accurate readings. The calibration process involves applying known weights or forces to the load cell and adjusting the output signal to match the expected values. Proper installation, protection from environmental factors like temperature and humidity, and periodic calibration are essential for reliable and accurate load cell measurements.[10]

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## HX711 Load Cell Amplifier Module

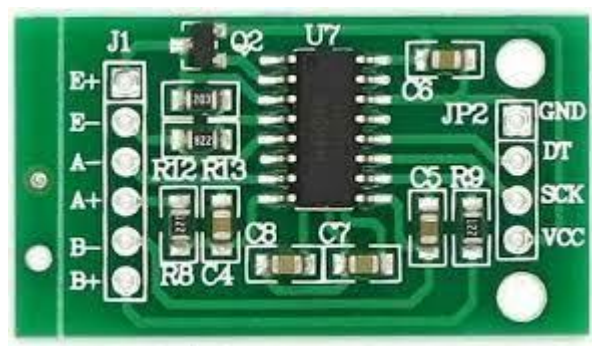


Fig 4.8 HX711 Load Cell Amplifier Module

The HX711 Load Cell Amplifier Module is a specialized electronic module that is designed to interface with load cells and enable them to be used with microcontrollers, such as Arduino or Raspberry Pi. It is commonly used in weighing applications, industrial automation, and other systems that require precise force or weight measurements.[11]

The HX711 module consists of an HX711 chip, which is a 24-bit analog-to-digital converter (ADC) that provides high-precision measurements of load cells. The module also includes an on-board amplifier and voltage regulator to ensure accurate and stable readings. The HX711 chip is capable of measuring load cells with a capacity of up to several hundred kilograms, with a resolution of up to 1 part in 16 million.

The HX711 module has two input channels for connecting load cells, and it communicates with the microcontroller through a digital interface, such as SPI or I2C. The module provides a digital output signal that can be easily processed by the microcontroller, which can then calculate the weight or force applied to the load cell based on the HX711's high-precision measurements.

The HX711 module is easy to use and requires minimal external components to operate. It is also highly configurable, with features such as adjustable gain, selectable sampling rates, and power-down modes to conserve power. The HX711 module can be used with a wide variety of load cells, making it a versatile and popular choice in many applications

In summary, the HX711 Load Cell Amplifier Module is a specialized electronic module that provides high-precision measurements of load cells, enabling them to be used with microcontrollers in a wide range of applications.[12]

- Breadboard
- Connecting wires

## 4.2 Software Components

Arduino IDE

Adafruit Industries IO

Adafruit MQTT Library

Adafruit GFX Library

Adafruit SSD1306 Library

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# **CHAPTER 5**

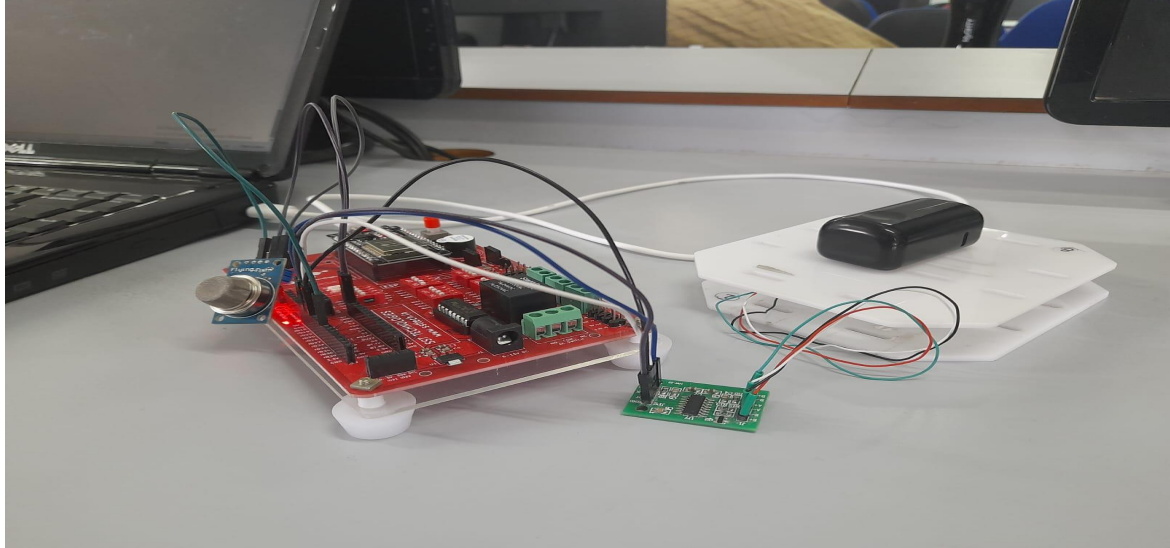
## **RESULTS**

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## CHAPTER 5

# RESULTS

### 5.1 Results



```
COM3
11:59:54.904 -> No Gas detected!
11:59:55.911 -> Load_cell output val: 33.19
11:59:56.918 -> No Gas detected!
11:59:57.926 -> Load_cell output val: 33.11
11:59:58.933 -> No Gas detected!
11:59:59.906 -> Load_cell output val: 32.99
12:00:00.915 -> No Gas detected!
12:00:01.922 -> Load_cell output val: 32.89
12:00:02.939 -> No Gas detected!
12:00:03.916 -> Load_cell output val: 32.81
12:00:04.930 -> No Gas detected!
12:00:05.906 -> Load_cell output val: 32.71
12:00:06.908 -> No Gas detected!
12:00:07.923 -> Load_cell output val: 32.61
12:00:08.929 -> No Gas detected!
12:00:09.935 -> Load_cell output val: 32.50
12:00:10.914 -> No Gas detected!
12:00:11.919 -> Load_cell output val: 32.42
12:00:12.931 -> No Gas detected!
12:00:13.906 -> Load_cell output val: 32.30
12:00:14.915 -> No Gas detected!
12:00:15.923 -> Load_cell output val: 32.23
12:00:16.903 -> No Gas detected!
12:00:17.910 -> Load_cell output val: 32.12
12:00:18.920 -> No Gas detected!
12:00:19.928 -> Load_cell output val: 32.00
12:00:20.902 -> No Gas detected!
```

Activate Windows  
Go to PC settings to activate Windows.

☒ Autoscroll ☒ Show timestamp

Newline 57600 baud Clear output



## 5.2 Advantages and Disadvantages

### Advantages

- Provides real-time gas leakage detection and gas quantity monitoring
- Alerts the user about gas leakage using real-time notifications
- Helps prevent accidents and ensures safety
- Can be remotely monitored using the cloud
- Easy to use and cost-effective

### Disadvantages

- Power requirements
- Integration challenges
- Limitations of the gas sensor
- Limited range of the load cell

## **CHAPTER 6**

### **CONCLUSION**

#### **CONCLUSION**

A cost-effective gas leakage detection system was proposed. Along with gas leakage detection, this system gives a fully automated approach towards the gas booking and safety precautions. Real time weight measurement of the gas and its display on LCD makes it an efficient home security system and also can be used in industries and other places to detect gas leaks. The cost involved in developing the system is significantly low and is much less than the cost of gas detectors commercially available in the market.

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